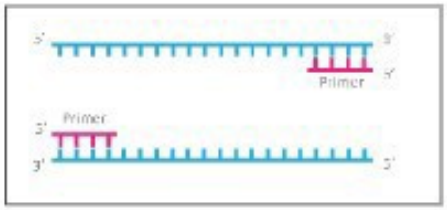
BT503 - Environment Biotechnology (Quiz 2) **Quiz Start Time: 06:1**

Question # 10 of 10 (Start time: 06:17:45 PM, 08 December 2022)



The figure above shows which step of PCR reaction?

Select the correct option

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○	Final Elongation
○	Denaturation
●	Annealing
○	Extension

Click to Save Answer & M

6:17 22.3 K/s 44
vulms.vu.edu.pk/ 8

BT503 - Environment Biotechnology (Quiz 2) Quiz Sta

Question # 9 of 10 (Start time: 06:16:22 PM, 08 December 2022)

The most famous Eukaryotic Vector is_____.

Select the correct option

- ○ Plasmid
- ◉ Virus
- ○ Yeast
- ○ Bacteria

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Click to Sav

6:23 ••• 13.3 K/s 46
Select text

major chemicals used in the manufacture of polyester fibres, films, adhesives, coatings and plastic bottles.

Methanogenic consortium of over 100 bacterial clones with the capability to digest terephthalate.

12.Mobility of DNA

Movement of genes within and between organisms. Genetic transfer through physical contact between bacteria (conjugation).Genetic transfer through bacteriophages (Transduction). Taking up of foreign DNA by bacteria (Transformation). Called extra-chromosomal elements (viruses and plasmids)

Rearrangement of genomic material within an organism stimulated by the presence of transposons. Move from one point and reinsert into a second site. Insertion may be into specific sites or random. Evolutionary process occurs principally by insertions and deletions of genomes caused by such activities. Such activities can also lead to rearrangement of existing structural genes into a different region of the genome. These genes then, therefore, operate under different parameters thus affecting the gene regulation.

e.g., Viruses which infect a wide host range, such as some retroviruses, the alfalfa mosaic virus and the Ti plasmid of *Agrobacterium tumefaciens*. Retroviruses have RNA as their genetic material. They replicate through double stranded DNA intermediate and thus can integrate into the host cell genome. RNA viruses tend to be more susceptible to mutation than DNA viruses presumably due to the less chemically stable nature of the macromolecule

A blurred distinction between chromosomal and extra-chromosomal elements (ECE). No organism lives in true genetic isolation as long as it is susceptible to at least one of ECE.

13.Genetic Manipulation

If selective breeding is to be considered as manipulation, then genes have been manipulated by man for a very long time.

Gregor Mendel.

Exchange of genetic information between organisms in nature is considerably more common than is generally imagined. In bacteria, the most likely candidates for genetic transfer are plasmids and bacteriophages.

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For eukaryotes, their most plausible vectors are eukaryotic viruses.

Exchange involving a vector requires compatibility between the organism donating the genetic material, the vector involved and the recipient organism. Microorganisms involved in remediation do so in their 'natural' state largely because they are indigenous at the site of contamination. In case of a sudden contamination, microbes are not able to adapt quickly. They may be 'trained' by artificial expansion of pre-existing pathways.

Or they can be genetically engineered.

Wild type and mutant.

Artificial mutations or deliberate reconstruction of the genome, often by the introduction of a novel gene. Basis of genetic engineering and has several advantages over traditional breeding techniques.

The process is specific i.e. only selected gene(s) is transferred.

14.The Manipulation of Bacteria Without Genetic Engineering

A general procedure is to take a sample of bacteria from, at, or near, the site of contamination from which a pure culture is obtained in the laboratory and identified, using standard microbiology techniques. Improve the bacterium's tolerance to the pollutant or to increase the capabilities of its pathways. Tolerance may be increased by culturing in growth medium containing increasing concentrations of the pollutant.

Catabolic expansion: Improving microbes ability to degrade a contaminant by culturing the bacteria in growth medium in which the contaminant is supplied as an essential part of the nutrition.

Only bacteria which have undergone a mutation enabling them to utilise this food source will be able to survive.

6:16 223 B/s 43
vulms.vu.edu.pk/ 8

BT503 - Environment Biotechnology (Quiz 2) Quiz Start Time

Question # 8 of 10 (Start time: 06:16:05 PM, 08 December 2022)

Reporter genes can only be a guide to the process(s) of _____ occurring in cell.

Select the correct option

- ○ Transcription
- ○ Translation
- ○ Replication
- ◉ Both B&C

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Click to Save Answ

□ ○ ◁

Question # 5 of 10 (Start time: 06:14:58 PM, 08 December 2022)

Transgenic plants?

Select the correct option

○	contain foreign genes in their cells
○	are used to produce human antibodies
◉	both (a) and (b) (Right answer)
○	are plants that differ in geographical locations

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Click to Save Answer &

BT503 - Environment Biotechnology (Quiz 2)

Quiz Start Time: 06:13 PM, 08 December

Question # 4 of 10 (Start time: 06:14:01 PM, 08 December 2022)

Total Mo

Plants may be genetically modified for the removal of toxic compounds from the environment. Many of the transgenic plants have been produced using the Ti plasmid transfer system of the bacterium _____ _____.

Select the correct option

- (●) Agrobacterium tumefaciencs
- ( ) Tobacco mosaic
- ( ) Thermotoga petrophila
- ( ) Acinetobacter baumannii

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Click to Save Answer & Move to Next Questio

BT503 - Environment Biotechnology (Quiz 2) **Quiz Start Time: 06**

Question # 1 of 10 (Start time: 06:13:30 PM, 08 December 2022)

Antibiotic resistance and β-galactosidase genes are _____ genes.

Select the correct option

- ○ illuminating
- ○ forecast
- ○ reluctant
- ◉ reporter

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6:07 15.0 K/s 39
vulms.vu.edu.pk/ 8

BT503 - Environment Biotechnology (Quiz 2) Quiz Start

Question # 10 of 10 (Start time: 06:06:49 PM, 08 December 2022)

Genomic libraries are very useful to look for specific sequence of?

Select the correct option

- All
- RNA
- DNA
- Protein

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6:06 688 B/s 39
vulms.vu.edu.pk/ 8

Time Left

BT503 - Environment Biotechnology (Quiz 2) Quiz Start Time: 06:02 PM, 08 Decemb

Question # 9 of 10 (Start time: 06:06:08 PM, 08 December 2022) Total

An organism containing a gene which doesn't belong to it and is derived from somewhere else then the organism is said to be _____

Select the correct option

- Mutant
- Transgenic
- transformed
- Modified

Click to Save Answer & Move to Next Ques

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6:06 297.4 K/s 38
vulms.vu.edu.pk/ 8

BT503 - Environment Biotechnology (Quiz 2)

Question # 8 of 10 (Start time: 06:05:52 PM, 08 December 2022)

Restriction enzymes are enzymes

Select the correct option

- ○ Capable of joining DNA molecules
- ○ Capable of restricting protein synthesis
- ○ Capable of adding nucleotide at 3'OH end
- ◉ Capable of cutting DNA molecule

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6:05 3.8 K/s 38
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BT503 - Environment Biotechnology (Quiz 2) Quiz

Question # 7 of 10 (Start time: 06:05:14 PM, 08 December 2022)

Enzyme used in PCR is_____.

Select the correct option

- ○ Restriction Endonuclease
- ○ None
- ◉ DNA polymerase
- ○ RNA polymerase

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□ ○ ◁

6:05 6.3 K/s 38
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BT503 - Environment Biotechnology (Quiz 2) Quiz Start Ti

Question # 6 of 10 (Start time: 06:04:10 PM, 08 December 2022)

Luciferase genes are also used at times for detection. Choose the correct statement for them.

Select the correct option

- ( ) Enzymes such as beta-galactosidase requires substrate X-gluc to produce light
- ( ) The detection requires provision of substrate which produces light
- ( ) Lucifearse genes are preferred over fluorescent proteins
- (•) They are obtained from fire flies only

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6:04 158 B/s 37
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BT503 - Environment Biotechnology (Quiz 2) Quiz

Question # 5 of 10 (Start time: 06:03:54 PM, 08 December 2022)

Which term explains cDNA best?

Select the correct option

- ○ Competent DNA
- ○ Compound DNA
- ○ Composed DNA
- ◉ Complementary DNA

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6:03 262 B/s 37
vulms.vu.edu.pk/ 8

BT503 - Environment Biotechnology (Quiz 2) Quiz

Question # 4 of 10 (Start time: 06:03:42 PM, 08 December 2022)

DNA is precipitated by_____.

Select the correct option

- CTAB
- Detergents
- Alcohol
- Sodium chloride

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Click to

6:03 195 B/s 37
vulms.vu.edu.pk/ 8

BT503 - Environment Biotechnology (Quiz 2) Quiz Start

Question # 3 of 10 (Start time: 06:03:28 PM, 08 December 2022)

Plasmids are used to

Select the correct option

- (●) carry genes into bacterial cells
- ( ) cut genes out of chromosomes
- ( ) make protein in bacterial cells
- ( ) join genes together

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□ ○ ◁

6:03 60.6 K/s 37
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BT503 - Environment Biotechnology (Quiz 2) Quiz Start Time: 06:0

Question # 2 of 10 (Start time: 06:02:22 PM, 08 December 2022)

Which of the following technique is most suitable for the detection of presence of gene product.

Select the correct option

- Dot blotting
- Western blotting
- Northern blotting
- Southern blotting

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6:02 2.1 K/s 37
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BT503 - Environment Biotechnology (Quiz 2) Quiz Start Tim

Question # 1 of 10 (Start time: 06:02:10 PM, 08 December 2022)

Baculovirus is a _____ virus.

Select the correct option

- fungal
- insect
- animal
- plant

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◻ ○ ◁

Which of the following is not a characteristic of a transgenic crop?

1) ◯ Bt insect resistance toxin

2) ◯ herbicide resistance

3) ◉ none of these

4) ◯ increased methionine content

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(Complaint Here As Incorrect)

3

5:37 vulms.vu.edu.pk/

BT503 - Environment Biotechnology (Quiz 2)

Question # 10 of 10 (Start time: 05:37:22 PM, 08 December 2022)

Yeast are _____.

Select the correct option

- (●) eukaryotic unicellular
- ( ) eukaryotic multicellular
- ( ) prokaryotic multicellular
- ( ) prokaryotic unicellular

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Click

5:37 ••• 756 B/s 26
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BT503 - Environment Biotechnology (Quiz 2)

Question # 9 of 10 (Start time: 05:36:39 PM, 08 December 2022)

Genomic libraries are very useful to look for specific sequence of?

Select the correct option

- RNA
- DNA
- All
- Protein

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Click

5:36 1.5 K/s 25
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BT503 - Environment Biotechnology (Quiz 2) Quiz

Question # 8 of 10 (Start time: 05:36:26 PM, 08 December 2022)

The process of expression of foreign genes in a plant is called_____

Select the correct option

- ○ Cell hybridization
- ○ Gene expression
- ◉ Genetic transformation
- ○ Transgenesis

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□ ○ ◁

5:35 118 B/s 25
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BT503 - Environment Biotechnology (Quiz 2)

Question # 6 of 10 (Start time: 05:35:02 PM, 08 December 2022)

Transgenic plants?

Select the correct option

- ○ are used to produce human antibodies
- ○ are plants that differ in geographical locations
- ● both (a) and (b) (Right answer)
- ○ contain foreign genes in their cells

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5:35 P 4G 0 B/s 25
vulms.vu.edu.pk/ 8

BT503 - Environment Biotechnology (Quiz 2) Qu

Question # 5 of 10 (Start time: 05:34:48 PM, 08 December 2022)

Which of the following technique is most suitable for the detection of RNA.

Select the correct option

- (●) Northern blotting
- ( ) Western blotting
- ( ) Dot blotting
- ( ) Southern blotting

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Click

□ ○ ◁

5:34 37 B/s 25
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BT503 - Environment Biotechnology (Quiz 2)

Question # 4 of 10 (Start time: 05:34:33 PM, 08 December 2022)

Restriction enzymes are enzymes

Select the correct option

- ○ Capable of joining DNA molecules
- ○ Capable of restricting protein synthesis
- ○ Capable of adding nucleotide at 3'OH end
- ● Capable of cutting DNA molecule

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5:34 5.0 K/s 25
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BT503 - Environment Biotechnology (Quiz 2) Quiz

Question # 3 of 10 (Start time: 05:34:03 PM, 08 December 2022)

Baculovirus is a _____ virus.

Select the correct option

- ○ animal
- ○ plant
- ○ fungal
- ◉ insect

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□ ○ ◁

5:34 156 B/s 24
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BT503 - Environment Biotechnology (Quiz 2) Quiz Start

Question # 2 of 10 (Start time: 05:33:50 PM, 08 December 2022)

A genetically modified organism means an organism which has had genes____.

Select the correct option

- (●) Replaced
- ( ) Divided
- ( ) Removed
- ( ) Added

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5:33 vulms.vu.edu.pk/

BT503 - Environment Biotechnology (Quiz 2)

Question # 1 of 10 (Start time: 05:33:28 PM, 08 December 2022)

Plasmids are used to

Select the correct option

- ○ cut genes out of chromosomes
- ○ make protein in bacterial cells
- ○ join genes together
- ◉ carry genes into bacterial cells

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5:31 43 B/s 23
vulms.vu.edu.pk/ 8

BT503 - Environment Biotechnology (Quiz 2) Qu

Question # 10 of 10 (Start time: 05:31:24 PM, 08 December 2022)

Which of the following statement is true?

Select the correct option

- A vector should have unique restriction sites
- All
- A vector should have an origin of replication
- A vector should have selectable markers

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5:31 1.6 K/s 23
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BT503 - Environment Biotechnology (Quiz 2) Qu

Question # 9 of 10 (Start time: 05:31:04 PM, 08 December 2022)

A genetically modified organism means an organism which has had genes____.

Select the correct option

- ○ Removed
- ◉ Replaced
- ○ Added
- ○ Divided

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5:31 8.0 K/s 23
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BT503 - Environment Biotechnology (Quiz 2)

Question # 8 of 10 (Start time: 05:30:25 PM, 08 December 2022)

Denaturation step in PCR usually occur at which temperature?

Select the correct option

- ○ 60◦C
- ◉ 95◦C
- ○ 20-30◦C
- ○ 72◦C

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5:30 P 4G 0 B/s 23
vulms.vu.edu.pk/ 8

BT503 - Environment Biotechnology (Quiz 2)

Question # 7 of 10 (Start time: 05:30:08 PM, 08 December 2022)

Restriction enzymes are enzymes

Select the correct option

- ○ Capable of adding nucleotide at 3'OH end
- ◉ Capable of cutting DNA molecule
- ○ Capable of restricting protein synthesis
- ○ Capable of joining DNA molecules

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5:30 209 B/s 23
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BT503 - Environment Biotechnology (Quiz 2) Quiz Start Time: 05:2

Question # 6 of 10 (Start time: 05:29:53 PM, 08 December 2022)

An organism whose genome has been altered by the gene transfer from another species is known as a_____-.

Select the correct option

- ○ Hybrid organism
- ○ Altered organism
- ○ New organism
- ◉ Transgenic organism

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5:29 88 B/s 23
vulms.vu.edu.pk/ 8

BT503 - Environment Biotechnology (Quiz 2)

Question # 5 of 10 (Start time: 05:28:55 PM, 08 December 2022)

Plasmids are used to

Select the correct option

- (●) carry genes into bacterial cells
- ( ) join genes together
- ( ) cut genes out of chromosomes
- ( ) make protein in bacterial cells

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Click

□ ○ ◁

5:28 ••• 4G 1.0 K/s 22
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BT503 - Environment Biotechnology (Quiz 2) Qu

Question # 4 of 10 (Start time: 05:28:39 PM, 08 December 2022)

Yeast are _____.

Select the correct option

- prokaryotic multicellular
- eukaryotic unicellular
- prokaryotic unicellular
- eukaryotic multicellular

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5:28 2.4 K/s 22
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BT503 - Environment Biotechnology (Quiz 2)

Question # 3 of 10 (Start time: 05:28:26 PM, 08 December 2022)

The process of expression of foreign genes in a plant is called_____

Select the correct option

- ○ Transgenesis
- ○ Cell hybridization
- ◉ Genetic transformation
- ○ Gene expression

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5:28 vulms.vu.edu.pk/

BT503 - Environment Biotechnology (Quiz 2)

Question # 2 of 10 (Start time: 05:28:12 PM, 08 December 2022)

The term often used as a synonym for Pre-mRNA is_____.

Select the correct option

- (●) hnRNA
- ( ) cDNA
- ( ) siRNA
- ( ) tnRNA

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5:25 244 B/s 21

Select text

permeable to small fragments of DNA and then the cells allowed to recover and grow into plants. These methods can be unsuccessful due to difficulties in recovery of the cells from the rather traumatic treatments and also because the DNA introduced, has a tendency to be inserted randomly

into the genome, rather than at a defined site. However, both methods enjoy the advantage that DNA enters the cell exactly as constructed and has not passed through an intermediate vector giving the opportunity for gene rearrangement.

84. Explain the initial steps of solid waste. 10

Ans:

1. Identify the types of waste.
2. Identify the sources of waste.
3. Determine the potential health hazards from waste.
4. Determine the volume of waste generated
5. Identify safe collection method/s.
6. Identify safe transportation method/s.
7. Identify safe disposal method/s.

85. Difference between expression and gene reporter 10

Expression vectors

These are similar to the vectors described above but in addition have the required signals located before and after the 'foreign' gene which direct the host cell to translate the product of transcription into a protein. It is sometimes a difficult, expensive or time-consuming procedure to analyse for product from the 'foreign' gene and so, in addition to the selector genes described above, there are frequently reporter genes to indicate whether or not the signals are 'switched on' allowing the 'foreign' DNA to be expressed. There are many reasons which are difficult to predict, why even a perfectly constructed gene may not be functional, such as the consequence of the exact site of insertion in the genome; hence the need for inbuilt controls.

Muhammad Kashif (03064969359)

Reporter Genes

There are many such genes in common use and these usually code for a trait or an enzyme. e.g., antibiotic resistance gene, or β-galactosidase, etc.

Blue/White screening.

Other reporter genes produce enzymes which can cause the emission of light such as the luciferase isolated from fireflies, or whose activity is easy and quick to assay like the bacterial β-glucuronidase (GUS) which is probably the most frequently used reporter genes in transgenic plants.

Reporter genes can only be a guide to the process of transcription and translation process. As with selector genes, the reporter genes serve no useful purpose once the cloning procedure has been successfully accomplished to produce the finished product.

86. How plant clean our environment and water body. 10

Plants improve air quality through several mechanisms: they absorb carbon dioxide and release oxygen through photosynthesis, they increase humidity by transpiring water vapor through microscopic leaf pores, and they can passively absorb pollutants on the external surfaces of leaves and on the plant root-soil system.

Planktons are profoundly sensitive to natural change and thus are best markers of water quality, especially lakes. Many planktons reproducing at an increased rate in lakes when there are high

5:19 424 B/s 19
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BT503 - Environment Biotechnology (Quiz 2)

Quiz Start Time: 05:12 PM, 08 Dec

Question # 9 of 10 (Start time: 05:19:02 PM, 08 December 2022)

To

A multiple cloning site (MCS), is also called a_____.

Select the correct option

Reload Math

- ( ) Operon
- ( ) promotor
- (•) .Restriction site
- ( ) Polylinker

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5:18 3.9 K/s 18
vulms.vu.edu.pk/ 8

BT503 - Environment Biotechnology (Quiz 2) Quiz Start Tim

Question # 8 of 10 (Start time: 05:18:13 PM, 08 December 2022)

Luciferase genes are also used at times for detection. Choose the correct statement for them.

Select the correct option

- ○ Enzymes such as beta-galactosidase requires substrate X-gluc to produce light
- ○ Lucifearse genes are preferred over fluorescent proteins
- ◉ They are obtained from fire flies only
- ○ The detection requires provision of substrate which produces light

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5:18 173 B/s 18
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BT503 - Environment Biotechnology (Quiz 2) Quiz Sta

Question # 7 of 10 (Start time: 05:17:12 PM, 08 December 2022)

Restriction enzymes are enzymes

Select the correct option

- ○ Capable of restricting protein synthesis
- ○ Capable of adding nucleotide at 3'OH end
- ◉ Capable of cutting DNA molecule
- ○ Capable of joining DNA molecules

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BT503 - Environment Biotechnology (Quiz 2)

Quiz Start Time: 05:12 PM, 08 Decem

Question # 6 of 10 (Start time: 05:16:25 PM, 08 December 2022)

Tota

Plants may be genetically modified for the removal of toxic compounds from the environment. Many of the transgenic plants have been produced using the Ti plasmid transfer system of the bacterium _____ _____.

Select the correct option

- Tobacco mosaic
- Acinetobacter baumannii
- Thermotoga petrophila
- Agrobacterium tumefaciencs

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5:16 ••• 4G 0 B/s 17
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BT503 - Environment Biotechnology (Quiz 2)

Question # 5 of 10 (Start time: 05:15:54 PM, 08 December 2022)

Which of the following technique is most suitable for the detection of RNA.

Select the correct option

- ○ Southern blotting
- ◉ Northern blotting
- ○ Western blotting
- ○ Dot blotting

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□ ○ ◁

5:15 78 B/s 17
vulms.vu.edu.pk/ 8

BT503 - Environment Biotechnology (Quiz 2) Quiz Star

Question # 4 of 10 (Start time: 05:15:16 PM, 08 December 2022)

Reporter genes can only be a guide to the process(s) of _____ occurring in cell.

Select the correct option

- ○ Translation
- ○ Replication
- ○ Transcription
- ◉ Both B&C

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BT503 - Environment Biotechnology (Quiz 2) Quiz Sta

Question # 3 of 10 (Start time: 05:13:43 PM, 08 December 2022)

Antibiotic resistance and β-galactosidase genes are _____ genes.

Select the correct option

- reluctant
- illuminating
- reporter
- forecast

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Click to Sav

□ ○ ◁

5:13 vulms.vu.edu.pk/

BT503 - Environment Biotechnology (Quiz 2)

Question # 2 of 10 (Start time: 05:12:40 PM, 08 December 2022)

Why is a eukaryotic cloning vector required?

Select the correct option

- ○ Cloning ease
- ○ variety
- ● large scale production
- ○ availability

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5:12 ••• 4G 17 B/s 16
vulms.vu.edu.pk/ 8

BT503 - Environment Biotechnology (Quiz 2)

Question # 1 of 10 (Start time: 05:12:13 PM, 08 December 2022)

Transgenic plants?

Select the correct option

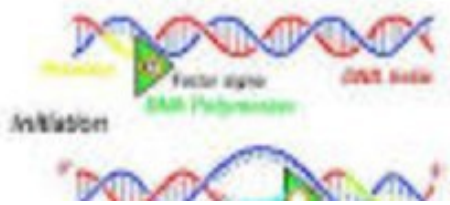
- ○ contain foreign genes in their cells
- ○ are plants that differ in geographical locations
- ◉ both (a) and (b) (Right answer)
- ○ are used to produce human antibodies

Muhammad Kashif (03064963933)

□ ○ ◁

The term **hnRNA** is often used as a synonym for pre-mRNA, although, in the strict sense

Muhammad Kashif (03064963933)



5:02 156.3 K/s 13
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BT503 - Environment Biotechnology (Quiz 2)

Question # 10 of 10 (Start time: 05:01:23 PM, 08 December 2022)

The process of expression of foreign genes in a plant is called_____

Select the correct option

- ○ Gene expression
- ○ Cell hybridization
- ○ Transgenesis
- ◉ Genetic transformation

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5:01 vulms.vu.edu.pk/

BT503 - Environment Biotechnology (Quiz 2)

Question # 9 of 10 (Start time: 05:00:14 PM, 08 December 2022)

A multiple cloning site (MCS), is also called a_____.

Select the correct option

○	Polylinker
●	.Restriction site
○	Operon
○	promotor

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5:00 168 B/s 12
vulms.vu.edu.pk/ 8

BT503 - Environment Biotechnology (Quiz 2) Quiz Start Time: 0

Question # 8 of 10 (Start time: 04:59:42 PM, 08 December 2022)

An organism whose genome has been altered by the gene transfer from another species is known as a_____

Select the correct option

- ○ Altered organism
- ◉ Transgenic organism
- ○ New organism
- ○ Hybrid organism

Click to Save Answer

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4:59 ••• 4G 1.1 K/s 12
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BT503 - Environment Biotechnology (Quiz 2) Qu

Question # 7 of 10 (Start time: 04:59:23 PM, 08 December 2022)

A plasmid can be considered as a suitable cloning vector if?

Select the correct option

- (●) All of the above
- ( ) it can be readily isolated from the cells
- ( ) insertion of foreign DNA does not alter its replication properties
- ( ) it possesses a single restriction site for one or more restriction enzymes

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BT503 - Environment Biotechnology (Quiz 2)

Question # 6 of 10 (Start time: 04:58:59 PM, 08 December 2022)

DNA is precipitated by_____.

Select the correct option

- ○ Sodium chloride
- ○ CTAB
- ◉ Alcohol
- ○ Detergents

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4:58 152 B/s 12
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BT503 - Environment Biotechnology (Quiz 2)

Question # 5 of 10 (Start time: 04:58:11 PM, 08 December 2022)

The sequence recognized by restriction enzyme is known as

Select the correct option

- (●) Recognition site
- ( ) Restriction site
- ( ) Both
- ( ) Cleavage site

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BT503 - Environment Biotechnology (Quiz 2) Quiz

Question # 4 of 10 (Start time: 04:57:40 PM, 08 December 2022)

Which of the following statement is true?

Select the correct option

- ○ A vector should have an origin of replication
- ○ A vector should have unique restriction sites
- ○ A vector should have selectable markers
- ◉ All

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BT503 - Environment Biotechnology (Quiz 2)

Question # 3 of 10 (Start time: 04:55:29 PM, 08 December 2022)

Which of the following technique is most suitable for the detection of RNA.

Select the correct option

- ○ Southern blotting
- ◉ Northern blotting
- ○ Dot blotting
- ○ Western blotting

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BT503 - Environment Biotechnology (Quiz 2)

Question # 2 of 10 (Start time: 04:55:11 PM, 08 December 2022)

Which term explains cDNA best?

Select the correct option

○	Compound DNA
○	Competent DNA
○	Composed DNA
◉	Complementary DNA

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4:55 223 B/s 10
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BT503 - Environment Biotechnology (Quiz 2)

Question # 1 of 10 (Start time: 04:53:41 PM, 08 December 2022)

The term often used as a synonym for Pre-mRNA is_____.

Select the correct option

- ○ tnRNA
- ◉ hnRNA
- ○ cDNA
- ○ siRNA

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BT406 - Research Methodology & Skill Enhancement (Quiz No. 1)

Question # 10 of 10 (Start time: 01:02:03 AM, 08 December 2022)

Difference between formative evaluation and summative evaluation is_____.

Select the correct option

- ○ Built in monitoring and final outcome report
- ○ Initial monitoring report and final outcome report
- ● Only final monitoring report
- ○ Built in monitoring and continuous monitoring

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BT503 - Environment Biotechnology (Quiz 2) Quiz St

Question # 1 of 10 (Start time: 06:42:34 PM, 08 December 2022)

Genomic libraries are very useful to look for specific sequence of?

Select the correct option

- (●) DNA
- ( ) Protein
- ( ) All
- ( ) RNA

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BT503 - Environment Biotechnology (Quiz 2) Quiz Start Time: 06:42 PM,

Question # 3 of 10 (Start time: 06:43:06 PM, 08 December 2022)

A plasmid can be considered as a suitable cloning vector if?

Select the correct option

- it can be readily isolated from the cells
- it possesses a single restriction site for one or more restriction enzymes
- insertion of foreign DNA does not alter its replication properties
- All of the above

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6:43 3.7 K/s 57
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BT503 - Environment Biotechnology (Quiz 2) Quiz Start Time: 06:4

Question # 4 of 10 (Start time: 06:43:25 PM, 08 December 2022)

In eukaryotes, the first product of transcription from DNA is

Select the correct option

- mRNA
- RNA
- Protein
- hnRNA

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6:44 139 B/s 57
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BT503 - Environment Biotechnology (Quiz 2) Quiz Start T

Question # 5 of 10 (Start time: 06:44:01 PM, 08 December 2022)

Enzyme used in PCR is_____.

Select the correct option

- RNA polymerase
- Restriction Endonuclease
- None
- DNA polymerase

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BT503 - Environment Biotechnology (Quiz 2) Quiz Start Time:

Question # 6 of 10 (Start time: 06:44:15 PM, 08 December 2022)

Why is a eukaryotic cloning vector required?

Select the correct option

- (●) large scale production
- ( ) Cloning ease
- ( ) availability
- ( ) variety

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6:46 P ••• 4G 734 B/s 58
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BT503 - Environment Biotechnology (Quiz 2) Quiz Start Tim

Question # 10 of 10 (Start time: 06:45:50 PM, 08 December 2022)

In molecular cloning technique Vector is a molecule used as a vehicle to carry_____.

Select the correct option

- ○ No DNA only RNA
- ○ Internal DNA
- ○ Non-coding DNA
- ◉ Foreign Genetic Material

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7:07 729 B/s 70
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BT503 - Environment Biotechnology (Quiz 2)

Question # 1 of 10 (Start time: 07:06:59 PM, 08 December 2022)

Which of the following is not a characteristic of a transgenic crop?

Select the correct option

- (●) None
- ( ) Increased methionine content
- ( ) Bt insect resistant toxin
- ( ) Herbicide resistance

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BT503 - Environment Biotechnology (Quiz 2)

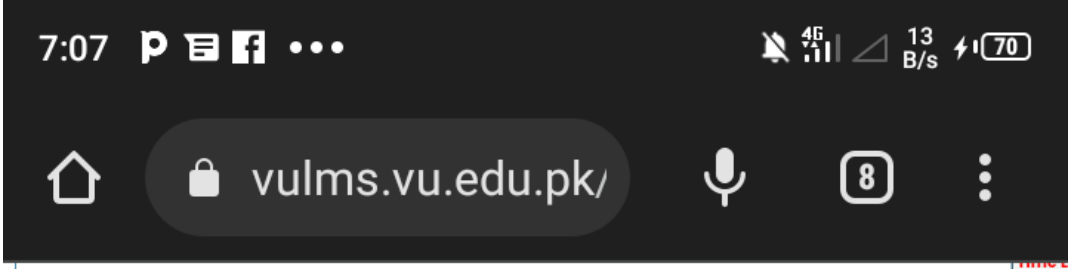
Question # 2 of 10 (Start time: 07:07:15 PM, 08 December 2022)

Restriction enzymes are enzymes

Select the correct option

- (●) Capable of cutting DNA molecule
- ( ) Capable of adding nucleotide at 3'OH end
- ( ) Capable of restricting protein synthesis
- ( ) Capable of joining DNA molecules

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BT503 - Environment Biotechnology (Quiz 2)

Quiz Start Time: 07:06 PM, 08 Dece

Question # 3 of 10 (Start time: 07:07:33 PM, 08 December 2022)

An organism whose genome has been altered by the gene transfer from another species is known as a_____-.

Select the correct option

Reload Math

- New organism
- Hybrid organism
- Transgenic organism
- Altered organism

Click to Save Answer & Move to Next

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BT503 - Environment Biotechnology (Quiz 2) Quiz Start Time: 07:

Question # 4 of 10 (Start time: 07:07:46 PM, 08 December 2022)

Denaturation step in PCR usually occur at which temperature?

Select the correct option

- (●) 95◦C
- ( ) 72◦C
- ( ) 20–30◦C
- ( ) 60◦C

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7:08 32 B/s 71
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BT503 - Environment Biotechnology (Quiz 2) Quiz Start Time: 07:06 PM, 08 D

Question # 5 of 10 (Start time: 07:08:00 PM, 08 December 2022)

The cloning vector may be DNA taken from a virus, the cell of a higher organism, or it may be the_____ of a bacterium.

Select the correct option

Reload M

- [ ] Capsule
- [x] Plasmid
- [ ] Cell Wall
- [ ] Chromosomal DNA

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BT503 - Environment Biotechnology (Quiz 2) Quiz Start

Question # 6 of 10 (Start time: 07:08:52 PM, 08 December 2022)

The most famous Eukaryotic Vector is_____.

Select the correct option

- ○ Bacteria
- ○ Plasmid
- ○ Yeast
- ◉ Virus

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7:09 26 B/s 71
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BT503 - Environment Biotechnology (Quiz 2) Quiz Start T

Question # 9 of 10 (Start time: 07:09:26 PM, 08 December 2022)

Plasmids are used to

Select the correct option

- (●) carry genes into bacterial cells
- ( ) join genes together
- ( ) cut genes out of chromosomes
- ( ) make protein in bacterial cells

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BT503 - Environment Biotechnology (Quiz 2) Quiz Start Tin

Question # 7 of 10 (Start time: 07:09:07 PM, 08 December 2022)

Yeast are _____.

Select the correct option

- [ ] prokaryotic unicellular
- [x] eukaryotic unicellular
- [ ] eukaryotic multicellular
- [ ] prokaryotic multicellular

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